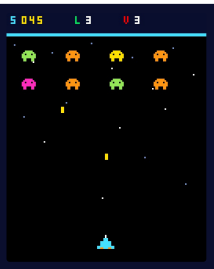
USER MANUAL

ROCKET GAME

ATmega328P Alien Shooter · Quick Start & Controls Guide

WHAT IS ROCKET GAME?

Rocket Game is a standalone, handheld arcade shooter built on a custom PCB. Move your rocket ship with the joystick, blast waves of aliens, and survive as many levels as you can. The whole game runs on a single microcontroller — just power it on and play.



In-game screen

CONTROLS

Joystick

Push in any direction to move your ship left, right, up or down within the play area. Release to stop — the ship will not drift.

Green Button — Start / Restart

Press to begin a new game from the title screen, or to restart after Game Over.

Red Button — Shoot

Press to fire at aliens. There's a brief reload delay between shots, so tap rather than mash.

HOW TO PLAY

1. **Power on.** Connect the board to a 5V power source via the USB-C port. The title screen appears automatically.
2. **Press the Green button** to start the game.
3. **Move your ship** with the joystick to dodge enemy fire and line up your shots.
4. **Press the Red button** to shoot the aliens above you. Each alien takes 1–3 hits depending on the level.
5. **Clear the wave.** Destroy every alien on screen to automatically advance to the next level.
6. **Survive.** You have 3 lives — getting hit by enemy fire or letting the aliens reach your row costs a life.
7. **Game Over?** Your final score is shown. Press the **Green button** to restart and try to beat it.

READING THE ON-SCREEN DISPLAY

SYMBOL	MEANING
S (cyan)	Your current score. Points increase with every alien destroyed — tougher levels are worth more.
L (green)	Current level (1–9). Aliens get faster, tougher, and shoot more often as the level rises.
V (red)	Lives remaining. The game ends when this reaches 0.

Alien Colour = Health

- **Red** — 3 hits to destroy
- **Orange** — 2 hits to destroy
- **Lime / Magenta / Yellow** — 1 hit to destroy

Sound Cues

- Short beep — you fired
- Medium beep — alien hit
- Long beep — you were hit
- Rising tone — level cleared
- Falling tone — game over

> **Tip:** The green and red buttons only do one job each — Green always starts or restarts, Red always shoots. If the screen looks frozen, check the power connection first before assuming the game has crashed.